

after the day's work is completed. In the final step, every month, backups are done manually by a systems engineer onto DVD media.

Hardware

File server, dual RAID 1 120 GB hard drives
One system with DVD writer and DVD media

Software

Personal backup managers
Professional backup application for servers

Humanware

One systems engineer

Medium-Sized Business

Company B has five network servers and needs to regularly back up between 80 and 250 GB of data. They generate or use over 80 GB of data a day, and need it all backed up and stored safely in case of a crash. This company relies solely on its data to function.

Individual employees use the various network servers to access work data. These SCSI network servers have a capacity of about 120 GB each, and have external hard drives connected to each server, taking hourly backups of each server's data. In addition, there is a NAS for redundancy and failure protection, which takes backups from the network servers at the end of each day. All this data is then backed up or archived on tape drive about once a week. The tapes are stored in a separate location, in order to enhance data security. This entire cycle of weekly full backups requires about five backup system engineers to monitor and maintain systems.

Hardware

Three file servers with dual RAID 1 SCSI hard drives
Tape drive with tape cassettes
NAS device

Software

Personal backup managers
Professional backup application for the servers

Humanware

Five systems engineers

Corporate

Company C is a corporate company with offices widely distributed across the globe. Let's say they have 10 worldwide offices, and all 10 need access to about 80 GB of core company data, apart from the backups of each office. The total amount of data that needs to be backed up here could be in the terabyte range. The company also needs the backups done every hour, and effectively works 24/7 due to the different time zones



Rough Estimates

Here's what you are looking at in terms of costs, for implementation of the backup solutions mentioned here.

Product	Type	Capacity	Rupees	Brands
Tape Servers	DAT	40 GB	21,750	Dell, Seagate
	DLT / Super DLT	80 GB	65,250	Quantum
	LTO	400 GB	174,000	HP
Tape Libraries	Super DLT, 8 slots	2.5 TB	174,000	Dell
	DLT, 100 slots	8 TB	282,750	Quantum
	LTO, 72 slots	28.8 TB	391,500	Dell
Tape Media	DAT (19 GB per hr)	40 GB	522	TDK / Maxell / Fuji
	DLT/SDLT (115 GB per hr)	320 GB	2,393	TDK / Maxell / Fuji
	AIT (112 GB per hr)	260 GB	2,393	TDK / Maxell / Fuji
	LTO (216 GB per hr)	400 GB	1,740	TDK / Maxell / Fuji
RAID Servers	Ultra ATA	1.2 TB	165,300	lomega, Dell, HP
NAS	SATA-150, Hot Swap	1 TB	282,750	lomega, Dell, HP

of its branch offices. The central backup with core company data is located in the head office.

Here, each individual office will have backup systems akin to medium-sized businesses, with the only difference being that specific data will be uploaded at pre-defined intervals to the head office servers. Each office will also have a NAS device as additional backup security.

If the core data is less than 20 GB, and daily uploads from each branch office are less than 300 MB, the company could opt for online storage or online backup. This would make it easier for the company to manage and share core data securely and easily. It also reduces the required infrastructure and manpower needed to monitor and maintain an online server.

The alternative is to locate the online server at the head office, under 24/7 monitoring by systems engineers. Here, each branch office needs about five engineers, and the head office needs to have systems maintenance engineers, security professionals, Web developers, etc.

Multiply the hardware and software requirements below by the number of branch offices you have.

Hardware

Five file servers
A NAS
Two Automated Tape drives
A T1 line
Online Storage

Software

Professional Distributed Backup Software

Humanware

A full-fledged IT department

Conclusion

These three models broadly categorise all medium to large-sized businesses across the globe. Your business may be exactly like one of these, or be somewhere between any two.

If you choose carefully, and wisely, you can use various combinations of the hardware and software mentioned here to find a perfect backup solution for your requirements. ■

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Backup or Archival?

People often do not understand the difference between backup and archival. Many small offices have a DVD-Writer as a backup solution for working data, and overspend on media. Due to these costs, backups are infrequent and loss of data may mean an unacceptable setback of days. On the other hand, some offices may have a server or external hard disk solution for archival purposes. This results in running out of hard disk space and in deletion of archives.

You need to identify whether your requirements are backup- or archival-based. A simple rule of thumb here is to use the previous self-assessment you did: if your backup requirements are infrequent and less than 5 GB, you are better off using DVD-R or DVD-RW media.